ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
COMPUTER SCIENCE & ENGINEERING
4th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Course Structure
(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Computer Science & Engineering
Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	CS241401	PCC	Database Management System	3	0	0	3	30	70
2	CS241402	PCC	Full Stack Application Development	2	0	0	2	0	50
3	CS241403	PCC	Machine Learning	3	0	0	3	30	70
4	CS241404	PCC	Computer Organization & Architecture	3	0	0	3	30	70
5	CS241405	PCC	Design and Analysis of Algorithms	3	0	0	3	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	CS241411	PCC	Database Management System Lab	0	0	2	1	15	35
9	CS241412	PCC	Full Stack Application Development Lab	0	0	4	2	15	35
10	CS241413	PCC	Java Programming Lab	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				**20**	**0**	**12**	**23**	**310**	**540**
Total Contact Hours per week: 32									
Total Credit: 23									

Course Code	Course Title	Hours per week L-T-P	Credit
CS241401	**Database Management System**	**3-0-0**	**3**

COURSE OUTCOMES

- **CO1** Understand database architecture, ER modeling, and relational model concepts. Understand (L2)
- **CO2** Apply relational algebra, relational calculus, and SQL to retrieve and manipulate data. Apply (L3)
- **CO3** Analyze functional dependencies and normalization techniques for relational schema design. Analyze (L4)
- **CO4** Evaluate transaction management and concurrency control mechanisms used in database systems. Evaluate (L5)
- **CO5** Apply NoSQL database techniques for managing unstructured and semi-structured data. Create (L3)

MODULE 1: Introduction to Databases and Data Modeling

Purpose of database systems, Characteristics of DBMS, Database users and architecture (3-level schema), Data models: Hierarchical, Network, Relational, ER model: entities, attributes, relationships, generalization, specialization, Constraints and ER-to-relational

MODULE 2: Relational Algebra, Relational Calculus, and SQL

Relational data model: concepts of relations, attributes, tuples, schemas, and keys, Relational Algebra: select, project, union, set difference, Cartesian product, joins, Tuple and domain relational calculus, SQL: DDL, DML, integrity constraints, SQL queries: joins, nested queries, aggregation, views, triggers.

MODULE 3: Database Design and Normalization

Functional dependencies and inference rules, Decomposition: lossless join and dependency preservation, Normal forms: 1NF, 2NF, 3NF, BCNF, 4NF, Multivalued dependencies, Schema refinement and normalization algorithms.

MODULE 4: Transactions and Concurrency Control

ACID properties, Transaction states, Schedules: serial, conflict-serializable, view-serializable, Concurrency control: two-phase locking, deadlock handling, Recovery techniques: log-based recovery, checkpoints, shadow paging.

MODULE 5: Query Processing and Indexing

Steps in query processing and optimization, Join algorithms: nested-loop, sort-merge, hash join, Indexing: primary, secondary, clustered and unclustered indexes, Index structures: B-trees and B+ trees, hashing techniques.

MODULE 6: Overview of Unstructured Data and NoSQL Databases

Structured vs. semi-structured vs. unstructured data, Introduction to NoSQL databases and the need for them, Brief introduction to different types of NoSQL: Document-based (e.g., MongoDB), Key-value stores (e.g., Redis), Columnar (e.g., Cassandra), Graph databases (e.g., Neo4j), CAP theorem and BASE properties, Data modeling and querying in MongoDB (CRUD and aggregations), Use cases in modern web and big data applications.

Text books/Reference books

1. **Abraham Silberschatz, Henry F. Korth, and S. Sudarshan**, Database System Concepts, McGraw-Hill Education (7th Edition or latest)
2. **Ramez Elmasri and Shamkant B. Navathe,** Fundamentals of Database Systems, Pearson Education
3. **Raghu Ramakrishnan and Johannes Gehrke,** Database Management Systems, McGraw-Hill.

4. **Pramod J. Sadalage & Martin Fowler,** NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Addison-Wesley.

Course Code	Course Title	Hours per week L-T-P	Credit
CS241402	**Full Stack Application Development**	**2-0-0**	**2**

Course Outcomes (COs):

- **CO1**: Understand the architecture and foundational concepts of full stack web development. Understand (L2)
- **CO2**: Apply frontend development tools and technologies to build responsive and dynamic interfaces. Apply (L3)
- **CO3**: Develop backend services and RESTful APIs with secure data handling. Apply (L3)
- **CO4**: Integrate frontend, backend, and databases to build full-stack web applications. Apply (L3)
- **CO5**: Demonstrate familiarity with deployment, version control, and CI/CD tools in real-world full stack systems. Apply (L3)

MODULE 1: Introduction to Web Technologies

Evolution from Web 1.0 to Web 3.0; Basics of Internet protocols: HTTP, HTTPS, DNS, IP addressing; Client-server architecture and request-response cycle; Overview of frontend, backend, and full-stack roles; Web browsers and developer tools

MODULE 2: HTML, CSS, and Responsive Web Design

HTML5: structure, semantic tags, forms, media; CSS3: selectors, box model, Flexbox, Grid; Responsive design with media queries; Introduction to CSS frameworks: Bootstrap / Tailwind CSS; Accessibility and web UI best practices

MODULE 3: JavaScript and Client-Side Scripting

JavaScript basics: variables, functions, objects, arrays; ES6+ features: arrow functions, let/const, promises, async/await; DOM manipulation and event handling; JSON structure and interchange format; Introduction to tools: npm, Babel, Webpack

MODULE 4: Modern Frontend Framework

Introduction to SPA frameworks (React / Vue / Angular); Components, Props, State, and Hooks (React); Routing using React Router / Vue Router; Fetching and displaying data from APIs

MODULE 5: Backend Development

Node.js fundamentals and Express.js framework; RESTful API development; Routing, middleware, error handling; Basics of authentication: JWT, OAuth (introductory)

MODULE 6: Database & Persistence

Comparison: Relational vs. Non-relational databases; MongoDB: Schema-less design, CRUD operations, Mongoose; SQL overview (MySQL/PostgreSQL - optional); Connecting database to Node.js applications; Security basics: SQL injection, CORS, CSRF

MODULE 7: Deployment & DevOps Essentials

Static & dynamic site hosting: Netlify, Vercel, Render; Version control using Git and GitHub; Environment variables and .env usage; CI/CD basics: GitHub Actions, build & deploy pipelines; Docker basics (optional)

Recommended Textbooks / References

1. Ethan Brown – *Web Development with Node and Express*, O'Reilly
2. Jon Duckett – *HTML & CSS: Design and Build Websites*, Wiley
3. Robin Nixon – *Learning PHP, MySQL & JavaScript*, O'Reilly
4. Alex Banks & Eve Porcello – *Learning React*, O'Reilly
5. MongoDB Manual – https://www.mongodb.com/docs/manual/

Course Code	Course Title	Hours per week L-T-P	Credit
CS241403	**Machine Learning**	**3-0-0**	**3**

Course Outcomes (COs):

- **CO1:** Explain the fundamental concepts of machine learning, including problem types, data representation, and learning paradigms. **Understand (Level 2) (Module 1)**
- **CO2:** Apply basic machine learning algorithms such as KNN, decision trees, and linear regression to solve classification and regression problems. **Apply (Level 3)** (**Module 2)**
- **CO3:** Analyze the performance and suitability of machine learning models using concepts like training-testing splits, overfitting, generalization, and validation techniques. **Analyze (Level 4)** (**Modules 2 & 3)**
- **CO4**: Evaluate and compare selected algorithms like Random Forests, SVM, Naïve Bayes, Logistic Regression, and K-means for various learning tasks. **Evaluate (Level 5) (Module 3)**
- **CO5:** Design and implement neural network-based solutions using MLPs, RNNs and CNNs, applying appropriate loss functions, optimization, and regularization techniques. **Create (Level 6) Module 4**

MODULE 1: Introduction to Machine learning
Motivation and role of machine learning in computer science and problem solving; Representation (features), linear transformations, Appreciate linear transformations and matrix vector operations in the context of data and representation; Problem formulations (classification and regression); Appreciate the probability distributions in the context of data, Prior probabilities and Bayes Rule; Introduce paradigms of Learning (primarily supervised and unsupervised. Also a brief overview of others)

MODULE 2: Fundamentals of Machine learning
PCA and Dimensionality Reduction; Nearest Neighbours and KNN; Linear Regression; Decision Tree Classifiers; Notion of Generalization and concern of Overfitting; Notion of Training, Validation and Testing; Connect to generalization and overfitting

MODULE 3: Selected Algorithms, Ensembling and RF; Linear SVM; K Means; Logistic Regression; Naive Bayes

MODULE 4: Neural Network based Learning
Role of Loss Functions and Optimization; Gradient Descent and Perceptron/Delta Learning; MLP, ; Backpropagation; MLP for Classification and Regression; Regularisation, Early Stopping; Introduction to Deep Learning; RNNs, CNNs

Text / Reference Books:

1. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, Mathematics for Machine Learning, Cambridge University Press (23 April 2020).
2. Tom M. Mitchell- Machine Learning - McGraw Hill Education, International Edition.
3. AurélienGéron Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, Inc. 2nd Edition.
4. Ian Goodfellow, YoshouaBengio, and Aaron Courville Deep Learning MIT Press Ltd, Illustrated edition.
5. Christopher M. Bishop Pattern Recognition and Machine Learning - Springer, 2nd edition.
6. Trevor Hastie, Robert Tibshirani, and Jerome Friedman - The Elements of Statistical Learning:
7. Data Mining, Inference, and Prediction - Springer, 2nd edition.

Course Code	Course Title	Hours per week L-T-P	Credit
CS241404	Computer Organization and Architecture	3-0-0	3

Course Outcomes (COs):

- **CO1: Explain** the fundamental components of a computer system and **illustrate** the instruction execution cycle and addressing modes. *(Understanding – Level 2) (Module 1)*
- **CO2: Analyze** the x86 architecture, CPU control unit design approaches, and various I/O management techniques to **evaluate** their impact on system performance. *(Analyzing – Level 4) (Module 2)*
- **CO3: Apply** memory organization techniques to **design** efficient memory management strategies. *(Applying – Level 3 & Creating – Level 6) (Module 3)*
- **CO4: Examine** the principles of pipelining and parallel processing, and **assess** their effects on system throughput and performance under different pipeline hazards. *(Evaluating – Level 5) (Module 4)*
- **CO5: Design** optimized computing architectures by integrating concepts of instruction set architecture, memory hierarchy, and parallel processing to **enhance** computational efficiency. *(Creating – Level 6) (All Modules)*

MODULE 1: Functional Components, Instruction Sets, and Data Representation

Functional blocks of a computer: CPU, memory, input-output subsystems, control unit; Instruction set architecture of a CPU registers; instruction execution cycle; RTL interpretation of instructions; addressing modes; instruction set; Case study: instruction sets of some common CPUs.
Data representation: signed number representation, fixed and floating point representations, character representation; Computer arithmetic: integer addition and subtraction, ripple carry adder, carry look-ahead adder, etc. multiplication shift-and add, Booth multiplier, carry save multiplier, etc. division restoring and non-restoring techniques, floating point arithmetic.

MODULE 2: x86 Architecture, CPU Control, Memory System, and I/O Management

Introduction to x86 architecture and instruction set; CPU control unit design: hardwired and micro-programmed design approaches, Case study design of a simple hypothetical CPU; Memory system design: semiconductor memory technologies, memory organization; Peripheral devices and their characteristics: Input-output subsystems, I/O device interface, I/O transfers—program controlled, interrupt driven and DMA; privileged and non-privileged instructions; software interrupts and exceptions; Programs and processes—role of interrupts in process state transitions, I/O device interfaces SCII, USB

MODULE 3: Memory Organization and Hierarchical Storage

Memory organization: Memory interleaving, concept of hierarchical memory organization, cache memory, cache size vs. block size, mapping functions, replacement algorithms, write policies.

MODULE 4: Pipelining and Parallel Processing

Pipelining: Basic concepts of pipelining, throughput and speedup, pipeline hazards; Parallel Processors: Introduction to parallel processors, Concurrent access to memory and cache coherency

Text / Reference Books:

1. "Computer Organization and Design: The Hardware/Software Interface", 5th Edition by David A. Patterson and John L. Hennessy, Elsevier.
2. "Computer Organization and Embedded Systems", 6th Edition by Carl Hamacher, McGraw Hill Higher Education.
3. "Computer Architecture and Organization", 3rd Edition by John P. Hayes, WCB/McGraw-Hill
4. "Computer Organization and Architecture: Designing for Performance", 10th Edition by William Stallings, Pearson Education.

5. "Computer System Design and Architecture", 2nd Edition by Vincent P. Heuring and Harry F. Jordan, Pearson Education.

Course Code	Course Title	Hours per week L-T-P	Credit
CS241405	**Design and Analysis of Algorithms**	**3-0-0**	**3**

Course Outcomes (COs):

- **CO1:** Analyze the time and space complexity of algorithms using asymptotic notations to assess the efficiency of algorithms. **(Analyzing – Level 4) (Module 1)**
- **CO2: Demonstrate** a familiarity with major algorithm design techniques **(Understanding – Level 2) (Modules 2)**
- **CO3:** Apply algorithmic strategies to solve classical problems like sorting and searching. **(Applying – Level 3) (Modules 3, 4, 5, and 6)**
- **CO4: Demonstrate** a familiarity with significance of some advanced topics, computational complexity classes and limits of computation **(Understanding – Level 2) (Modules 7)**
- **CO5:** Choose a design strategy and propose efficient algorithms to solve computational problems **(Creating – Level 7) (Modules 1,2,3,4,5,6 and 7)**

MODULE 1

Revision of data structure and algorithms; Notion of Algorithm; Time and space tradeoffs; Asymptotic Notations; Mathematical analysis of non-recursive algorithms; Mathematical analysis of recursive algorithm through recurrence relations: Substitution method, Recursion tree method and Masters' theorem.

MODULE 2

Introduction to Basic strategies: Brute-Force, Divide and Conquer, Greedy, Dynamic Programming, Branch and-Bound and Backtracking strategies; Notion of using these techniques for Problem-Solving.

MODULE 3

Examples and analysis of following: Selection sort, Bubble sort, Sequential searching (Linear Search), Brute force string matching, Merge sort, Quick Sort, Binary Search.

MODULE 4

Examples and analysis of following: Fractional Knapsack problem, Minimum cost spanning tree: Prim’s and Kruskal’s algorithm, Single source shortest path problem, Principle of optimality, all pair shortest path problem, 0/1 Knapsack problem, Traveling salesperson problem.

MODULE 5

Examples and analysis of following: General method backtracking, N-Queen problem, 0/1 Knapsack problem, General method of branch & bound, 0/1 Knapsack problem, Traveling sales person problem, Order Statistics and Median Selection.

MODULE 6

Graph and Analysis of Graph Traversal algorithms: Depth First Search and Breadth-first search algorithms; **Flow Algorithms:** Max Flow, Ford-Fulkerson, Applications; Disjoint Set Data Structure (Union-Find) and its applications in graph algorithms.

MODULE 7

Computability of Algorithms, computational complexity classes — P, NP, NP-complete and NP-hard. Cook's theorem, Classes of problems beyond NP- PSPACE, Reduction techniques, notion of approximation algorithms and randomized algorithm with simple examples.

Text / Reference Books:

1. Introduction to Algorithms, 4TH Edition,Thomas H Cormen, Charles E Lieserson, RonaldL Rivest and Clifford Stein, MIT Press/McGraw-Hill.
2. Fundamentals of Algorithms — E. Horowitz et al.
3. Algorithm Design, IST Edition, Jon Kleinberg and Eva Tardos, Pearson.

4. Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Michael T Goodrich and Roberto Tamassia, Wiley.
5. Algorithms a Creative Approach, 3RD Edition, Udi Manber, Addison-Wesley, Reading, MA.

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	**Finance and Accounting**	**3-0-0**	**3**

Prerequisites: High School Education

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash
- books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting

Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation

Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank
Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts

Concept of Capital Expenditure and Revenue Expenditure, Bad Debts,
Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment

Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market-Initial Public Offering, Market Players, Secondary Market: Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns- (Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	**Environmental Science**	**3-0-0**	**0**

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1**: Relate environment and its associated problems
- **CO2**: Explain basic ethics of living and protection of other organisms
- **CO3**: Applying sustainable development principles

MODULE 1: Introduction:
Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse
Land use: Impact of land–use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution
Introduction: Water quality standards, Water pollution

MODULE 4:
Air Pollution: Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5:
Noise Pollution: Introduction: Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6:
Control of pollutions: Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
CS241411	**Database Management System Lab**	**0-0-2**	**1**

Course Outcomes

- **CO1**: Implement SQL commands to create and manipulate relational database structures. Apply (L3)
- **CO2**: Develop and execute queries using relational algebra and advanced SQL features. Apply (L3)
- **CO3**: Analyze and normalize relational schemas using functional dependencies. Analyze (L4)
- **CO4**: Apply indexing and transaction control mechanisms to manage data consistency and performance. Apply (L3)
- **CO5**: Design and query NoSQL databases for managing unstructured or semi-structured data. Create (L6)

List of Experiments

Basic SQL – DDL, DML, DQL (CO1)

1. Create tables with primary and foreign key constraints using DDL commands.
2. Perform INSERT, UPDATE, DELETE operations using DML commands.
3. Retrieve data using SELECT with WHERE, ORDER BY, and GROUP BY clauses.
4. Use SQL aggregate functions: COUNT, SUM, AVG, MIN, MAX.

Advanced SQL – Joins, Views, Nested Queries (CO2)

5. Write queries using various types of JOINs (INNER, OUTER, SELF).
6. Implement nested queries and correlated subqueries.
7. Create and query VIEWS.
8. Define and enforce constraints (CHECK, NOT NULL, UNIQUE).

ER Design & Normalization (CO3)

9. Draw an ER diagram for a given scenario and convert it into relational schema.
10. Identify functional dependencies and apply 1NF, 2NF, 3NF.
11. Apply BCNF decomposition to a relational schema.

Transactions, Indexing, Recovery (CO4)

12. Simulate transactions using COMMIT, ROLLBACK, and SAVEPOINT.
13. Implement basic concurrency control using locking mechanism (simulated).
14. Simulate log-based recovery (theoretical or pseudo-code).
15. Create indexes on a table and measure performance differences with and without indexing.

NoSQL using MongoDB (CO5)

16. Install MongoDB and create a database and collection.
17. Perform CRUD operations on MongoDB documents.
18. Use MongoDB aggregation operations (match, group, sort).
19. Model nested/embedded documents to represent relationships.
20. Import JSON data into MongoDB and perform queries using Mongo Shell or Compass.

Additional Experiments: Additional experiments (if any) considered appropriate by the department may be conducted.

Text / Reference Books:

1. **Silberschatz, Korth, and Sudarshan,** Database System Concepts, 7th Edition, McGraw-Hill Education
2. **Ramez Elmasri and Shamkant B. Navathe**, Fundamentals of Database Systems, Pearson Education
3. **Raghu Ramakrishnan and Johannes Gehrke**, Database Management Systems, McGraw-Hill

4. **Pramod J. Sadalage and Martin Fowler**, NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Addison-Wesley

Course Code	Course Title	Hours per week L-T-P	Credit
CS241412	**Full Stack Application Development Lab**	**0-0-4**	**2**

Course Outcome

- CO 1: Create structured and responsive web pages using HTML, CSS, and layout frameworks. Create (L6)
- CO 2: Develop dynamic front-end behavior using JavaScript and handle user events and validation. Apply (L3)
- CO 3: Build interactive frontend applications using modern frameworks like React Apply (L3)
- CO4: Implement RESTful APIs and backend logic using Node.js and Express.js. Apply (L3)
- CO 5: Integrate frontend, backend, and database functionality and deploy full stack applications online. Apply (L3)

List of Experiments

No.	Experiment Title	Mapped CO
1	Create a static personal portfolio page using HTML5.	CO1
2	Design responsive layouts using CSS3 Flexbox and Grid.	CO1
3	Create a web form with various input types and validate structure using HTML5 features.	CO1
4	Use Bootstrap or Tailwind CSS to design a responsive navbar and footer.	CO1
5	Write JavaScript to manipulate DOM and handle button click events.	CO2
6	Implement form validation using JavaScript (regex for email, password).	CO2
7	Use ES6+ features like arrow functions and async/await in frontend code.	CO2
8	Fetch and display JSON data from a public API using JavaScript fetch API.	CO2
9	Build a simple React/Vue app with components, props, and state management.	CO3
10	Implement routing using React Router or Vue Router.	CO3
11	Create a weather app or similar using external REST API in a frontend framework.	CO3
12	Setup Node.js and create a basic Express.js server with routing.	CO4
13	Develop a RESTful API with GET and POST endpoints using Express.js.	CO4
14	Connect Express.js with MongoDB (using Mongoose) to store and retrieve data.	CO4
15	Implement authentication using JWT tokens (login/signup simulation).	CO4
16	Create a full-stack CRUD application (e.g., task manager or blog).	CO5
17	Integrate frontend (React/Vue) with backend (Node.js) using REST APIs.	CO5
18	Use environment variables and .env files securely in both frontend and backend.	CO5
19	Host a static site	CO5
20	Deploy a full-stack web app (frontend + backend + DB)	CO5

Additional Experiments: Additional experiments (if any) considered appropriate by the department may be conducted.

Course Code	Course Title	Hours per week L-T-P	Credit
CS241413	**Java Programming Lab**	**0-0-2**	**1**

Course Outcomes

- **CO1:** Implement basic Java programming concepts including control structures, classes, objects, inheritance, interfaces, and collections. *(Applying – Level 3)*
- **CO2:** Design user interfaces using AWT and Swing components along with event-driven programming for interactive Java applications. *(Creating – Level 6)*
- **CO3:** Develop multithreaded applications and implement basic client-server communication using Java networking concepts. *(Applying – Level 3)*
- **CO4:** Integrate Java programs with databases using JDBC for performing data manipulation and navigation. *(Applying – Level 3)*

MODULE 1

Implementing the basics concepts of Java language, control structures, classes and objects, inheritance, interfaces, collections

MODULE 2

Implementing User Interfaces in Java — JAVA AWT Package, Basics User Interface Components (Labels, buttons, Check boxes, Radio buttons, choice Menu or Choice Lists, Text fields, Text areas, scrolling list, scroll bars, panels and frames), Layouts(Flow, Grid, Border, Card),event-driven programmingevent driven programs, event handling process, Java's event types, JAVA Swings-Comparison between Swing and AWT, Java swing packages, Swing basic containers, Swing components, event handling using Java swing, using dialogs, Joptionpane class, input dialog boxes, Timers and Sliders, Tables, Borders for components.

MODULE 3

Implementing Threads in Java, Client server application, connecting to the Web

MODULE 4

Familiarity to implementing JAVA database connectivity, JDBC ODBC Bridge, JAVA.SQL package, connecting to remote data base. Only connecting methods are to be introduced (without prior knowledge of DBMS)

Experiments: Experiments are to be conducted with Java programming language. A set of experiments considered appropriate by the department may be conducted for each module.

Text / Reference Books:

1. Deitel&Deitel, JAVA: How to Program, Pearson education
2. Deitel&Deitel, Internet and World Wide Web How to Program, Pearson education
3. Ivan Bay Ross, Web Enabled Commercial Application using Java 2, BPB publication (1998)
4. David Flanagan, Java Script the Definitive Guide, O"relly, 5e (2006)

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	**Micro Project (Skill Based)**	**0-0-4**	**2**

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- CO1: Analyze engineering problems and their solutions
- CO2: Apply modern engineering tools to attempt solutions to the problems
- CO3: Adapt with computational thinking and automation skills
- CO4: Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5

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